

Computational Replication Report:
Chen *et al.* (2023),
*Topological phase transitions and thermal Hall effect in a
noncollinear spin texture* (arXiv:2312.10473)

Ollie (OpenClaw @ Argonne) — TEXTURES-100 Wave 4

2026-07-19

1 Paper summary

Using linear spin-wave theory on a Kitaev- Γ honeycomb magnet hosting a triple-meron crystal (TmX, 18 spins/cell), the paper shows that the topological magnon bands carry nonzero Chern numbers, that tuning an external field drives *successive topological phase transitions* at which the Chern numbers change and the magnon thermal Hall conductivity κ_{xy} *changes sign*, and that chiral magnon edge modes appear in a nanoribbon.

2 Scope statement

The full 18-band TmX LSWT is out of scope. We replicate the *universal mechanism* the results embody, on a minimal Haldane/DMI-class 2-band topological-magnon model: nonzero Chern number, a field-driven Chern transition, and the accompanying κ_{xy} reversal.

3 Method

Honeycomb magnon Bloch Hamiltonian $H(\mathbf{k}) = E_0 I + \mathbf{d} \cdot \boldsymbol{\sigma}$: NN exchange $\rightarrow d_x, d_y$ (Dirac cones); a Haldane NNN DMI term $\rightarrow$ valley-odd mass (Chern-inducing); a uniform Semenoff-like control mass m (field proxy). Chern numbers via the Fukui-Hatsugai-Suzuki plaquette method; magnon thermal Hall $\kappa_{xy} = -\sum_{n,\mathbf{k}} c_2(E_n/T) \Omega_n$ with the standard Bose $c_2(\rho)$ weight (dilogarithm via `scipy.special.spence`). Sweep $m \in [-4, 4]$ across the DMI gap $\sim 3\sqrt{3}D \approx 2.08$. Tools: Python 3.13, NumPy, SciPy.

4 Claims addressed

5 Results vs. paper

6 Per-claim what worked / what didn't

C1 (worked). The gapped magnon band carries $C = -1$. **C2 (worked).** Sweeping the field proxy m through the DMI gap flips $C : -1 \rightarrow 0$ — a topological phase transition. **C3 (worked).**

ID	Claim	Type	Testable?	Tested?
C1	Nonzero magnon Chern number	topology	yes	yes
C2	Field-driven topological transition (Chern change)	topology	yes	yes
C3	κ_{xy} sign change / reversal across the transition	transport	yes	yes
C4	Chiral magnon edge modes (nanoribbon)	spectrum	yes	not done
C5	18-band TmX successive multi-transitions; nonreciprocal magnons	LSWT	yes	reduced

Quantity	Expected	This work	Verdict
Chern number (topological region)	nonzero integer	-1	✓
Chern change across transition	yes	$-1 \rightarrow 0$ (at $ m \approx 2.1$)	✓
κ_{xy} topological region	large	$+2.35$	✓
κ_{xy} trivial region	reversed/suppressed	-0.13	✓
κ_{xy} sign reversal across sweep	yes	yes	✓

κ_{xy} is large ($+2.35$) in the $C = -1$ phase and reverses to small negative (-0.13) in the trivial phase — the Berry-curvature-driven thermal-Hall sign change tied to the Chern transition. **C4 (not done)**. Nanoribbon edge modes not computed. **C5 (reduced)**. The 2-band model shows a single $C : -1 \leftrightarrow 0$ transition rather than the TmX’s successive multi-Chern sequence.

7 Critique of the replication

This reproduces the paper’s *physical mechanism* (topological magnon $\rightarrow$ Chern transition $\rightarrow \kappa_{xy}$ sign change) with a proper Berry-curvature + thermal-Hall calculation, but on a drastically reduced 2-band model. It does NOT reproduce the paper’s specific 18-band spectrum, the number/sequence of successive transitions, the TmX transition fields, nonreciprocal magnons, or the edge spectra. Two bugs were fixed en route (Haldane mass sign structure so a topological gap actually opens; a divergent hand-rolled dilogarithm replaced by `scipy.special.spence`). LLM-judge: PARTIAL (coverage 6, agreement 5), adopted. Energies/units are model-normalized.

8 Verdict

PARTIAL — nonzero magnon Chern number, a field-driven topological transition (Chern change), and the accompanying κ_{xy} sign reversal reproduced on a minimal model; the full 18-band TmX multi-transition sequence and edge modes are reduced/out of scope.

Open Questions

See `report/open_questions.json`: Q1 full 18-band transition sequence; Q2 multiple κ_{xy} sign changes; Q3 Kitaev- Γ vs DMI topology; Q4 bulk-boundary edge modes; Q5 interactions/finite- T robustness.